

SARTHAK KHANDELWAL

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With 4 years of industry software development experience and a Master's degree in Computer Science, I bring expertise in Cloud Computing, Machine Learning, and Full Stack Development to contribute effectively to dynamic teams.

SKILLS

Skills: Python, AWS, Web Development, SQL, Machine Learning, Deep Learning, React, Spring Boot, Hibernate, Microservice, Computer Vision, Agile, LLM, Langchain, Vector DB, NLP, Go, GoLang, Poetry, CI/CD, BitBucket, Gitlab, Jenkins, Websockets, Twilio, Apache Kafka, PyTorch, TensorFlow

Languages: Python, GO, Javascript, HTML, CSS, JAVA

Framework and Tools: Postgres, AWS Serverless Framework, MySQL, Vuejs, MongoDB, AWS S3, React, Flask, Kubernetes, Docker, FastAPI, Django

WORK EXPERIENCE

University at Buffalo, Research Engineer

Feb 2024 – Jan 2025

3D Reconstruction Pipeline for Heritage Site Modeling (Python, PyTorch, AWS EC2, Docker, GitLab)

- Developed a **3D reconstruction pipeline** with **Gaussian splatting** and **Pytorch** for heritage site modelling
- Migrated on-premises pipeline to **AWS EC2**, built **Gitlab/Docker/Terraform CI/CD pipeline**, cutting **costs** by **50%**
- Created scalable image processing workflow with **AWS S3/Batch** managing **1,000+** crowdsourced **images** per site

Bajaj Finserv, Software Developer

Jan 2022 – Jan 2023

LOKAL (Location Based Dealer Finder) (Python, SQL, MongoDB, Flask, Docker, Kubernetes, Jenkins)

- Reduced the latency** of the geolocation-based **microservice**, optimizing dealer availability within specified distances
- Migrated spatial querying from an in-memory **KDTree** to geospatial indexing provided by **SQL** and **MongoDB** databases
- Developed a robust backend architecture using **Flask, SQL** and **MongoDB**. **Increased** service responsiveness by **20%**
- Streamlined **CI/CD** pipelines using **Jenkins**. Managed **Python** dependencies with **Poetry** and ensured code integrity via **Bitbucket**
- Deployed the service using **Docker** and orchestrated containers with **Kubernetes**

Retrieval-Augmented Generation (RAG) for Design Documents (Python, Docker, VectorDB, AWS S3, AWS Lambda)

- Implemented a **LangChain**-based RAG tool with **LLMs** for role-based access control and retrieval of **1000+** design **documents**
- Reduced** employee document **search time** by up to **50%** with an LLM-powered chatbot for fast, accurate retrieval
- Built a microservice to ingest, store, and manage **1000+** design documents using **S3, Lambda, Python**, and **Pinecone**
- Deployed the system using **Terraform, Docker, Kubernetes**, and **Jenkins** with **CI/CD pipelines** for scalable and reliable operations

Speaker Identification (Python, Docker, Kubernetes, Flask, Keras, TensorFlow)

- Reduced** transcript processing **time** from **2 hours** to **30 minutes** for public company earnings calls
- Developed a Python based pipeline using **TensorFlow, Flask** and **Amazon Transcribe** to transcribe and analyse audio data
- Built and deployed a **Docker** containerized speaker identification model on an **EC2** instance
- Integrated the inference service with **Apache Kafka** to consume transcription events and publish predictions
- Automated **CI/CD** pipelines using **Jenkins** and **Terraform**, enhancing system reliability

HighRadius, Software Developer Engineer, Intern

Aug 2021 – Jan 2022

- Migrated data and transformed queries from **SQL** to **Snowflake**

Iotron Technologies, Software Developer

Jan 2020 – June 2021

- Built responsive Art Gallery website with **React/Javascript**, integrated **Razorpay** payment gateway and **ShipRocket** shipping service

PROJECTS

Smart Assist: AI-Powered Call Management (GitHub)

- Developed an **AI assistant** that manages inbound call, manages appointments, and routes legitimate calls while filtering spam.
- Engineered using **FastAPI, WebSockets, Twilio Voice API, OpenAI's streaming API**, and **Google Calendar**.
- Implemented **asynchronous architecture** with concurrent **WebSocket streams**, **real-time audio processing**, and **SMS notification system**

EDUCATION

University at Buffalo SUNY

Master's in Computer Science and Engineering

CGPA: 3.8/4.0

SRM Institute of Science and Technology

Bachelor of Technology, Computer Science and Engineering

CGPA: 9.7/10.0

CERTIFICATIONS

Python Bootcamp | Full Stack Web Development (HTML|CSS|JavaScript) | Cloud Computing (NPTEL) | Deep Learning Specialization